

Math Camp Exam Solutions 2020

1.

(a) True. Let us denote $S = \{x_1, x_2, \dots, x_n\} \subset X$.

Closed: S is closed iff $X \setminus S$ is open. Let $x \in X \setminus S$ and $\varepsilon = \varepsilon(x) = \min_{1 \leq i \leq n} d(x, x_i) > 0$. By construction, it holds

$$x' \in B_\varepsilon(x) \implies d(x, x') < \min_{1 \leq i \leq n} d(x, x_i) \implies x' \notin S.$$

This shows $B_\varepsilon(x) \subset X \setminus S$. Since $x \in X \setminus S$ is arbitrary, $X \setminus S$ is open, thereby showing S is closed.

Bounded: Since it holds that

$$x \in S \implies d(x, x_1) < r := \max_{1 \leq i \leq n} d(x_i, x_1) + 1 \in (0, \infty),$$

$S \subset B_r(x_1)$ holds, i.e., S is bounded.

Compact: Let $\mathcal{U} = \{U_i : i \in I\}$ be an open cover of S . Then for each $x_k \in S$, $k \leq n$, there is $i(k) \in I$ such that $x_k \in U_{i(k)}$. It follows that $\{U_{i(k)} : k \leq n\}$ is a finite subcover of $\mathcal{U}$. Thus S is compact.

Note you can also just show compactness and argue compact implies closed and bounded.

(b) True. Let $S = \{\vec{v}_i : i \in I\} \cup \{\vec{0}\}$ be such a collection. Since

$$\sum_{i \in I} 0 \cdot \vec{v}_i + 1 \cdot \vec{0} = \vec{0},$$

there is a non-trivial linear combination in S that is equal to $\vec{0}$. Thus S is not linearly independent.

(c) False. Let us define $f(x) = e^{2x}$ and $g(x) = -2e^x + 1$ for $x \in \mathbb{R}$. Any (weakly) increasing or decreasing function $h : \mathbb{R} \rightarrow \mathbb{R}$ is quasiconcave because for all $x, y \in \mathbb{R}$ and $\alpha \in (0, 1)$, it holds (note that $\min\{x, y\} \leq \alpha x + (1 - \alpha)y \leq \max\{x, y\}$)

$$\begin{aligned} h(\alpha x + (1 - \alpha)y) &\geq h(\min\{x, y\}) = \min\{h(x), h(y)\} \text{ if } h \text{ is increasing,} \\ h(\alpha x + (1 - \alpha)y) &\geq h(\max\{x, y\}) = \min\{h(x), h(y)\} \text{ if } h \text{ is decreasing.} \end{aligned}$$

(Recall that quasiconcavity and quasiconvexity are preserved under monotone transformations. h and $(-h)$ act as a monotone transformation of $x \mapsto x$ when h is increasing and decreasing, respectively.) Since $f'(x) = 2e^{2x} > 0$ and $g'(x) = -2e^x < 0$, this implies that f and g are quasiconcave on $\mathbb{R}$. However, $f(x) + g(x) = e^{2x} - 2e^x + 1 = (e^x - 1)^2$ is not quasiconcave since the upper contour set

$$\{x \in \mathbb{R} : f(x) + g(x) \geq 1/4\} = \{x : e^x \leq 1/2\} \cup \{x : e^x \geq 3/2\} = (-\infty, \log(1/2)] \cup [\log(3/2), \infty)$$

is not convex.

The above argument can be generalized to give a multi-dimensional counterexample in $\mathbb{R}^d$, $d \geq 2$. For instance, when $d = 2$, $f(x, y) = e^{2(x+y)}$ and $g(x, y) = -2e^{x+y} + 1$ are quasiconcave as monotone transformations of linear functions. However, $f + g$ lacks quasiconcavity by the same logic.

(d) True. Let $\{K_i : i \in I\}$ be a non-empty collection of compact sets. Heine-Borel theorem dictates that each K_i is closed and bounded in $\mathbb{R}$. Thus, the intersection $\bigcap_{i \in I} K_i$ of closed sets is also closed in $\mathbb{R}$. $\bigcap_{i \in I} K_i$ is also bounded since it is a subset of a bounded set K_{i_0} for some $i_0 \in I$. By Heine-Borel theorem, $\bigcap_{i \in I} K_i$ is also compact. Likewise, the finite union of closed and bounded sets is also closed and bounded and hence compact (Heine-Borel).

(e) False. A counterexample: $f : (\mathbb{R}, d_2) \rightarrow (\mathbb{R}, d_2)$, $f(x) = \sin x$, $x \in \mathbb{R}$. Then f is a continuous function and $(0, 2\pi)$ is open in $(X, d_X) = (\mathbb{R}, d_2)$, but $f((0, 2\pi)) = [-1, 1]$ is not open in $(Y, d_Y) = (\mathbb{R}, d_2)$.

(f) False. $O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ is a diagonal matrix, which is clearly diagonalizable. But O is not invertible.

(g) True. Assume to the contrary $x > b$ and let $\varepsilon = x - b > 0$. There exists $N \geq 1$ such that

$$n \geq N \implies |x_n - x| < \varepsilon/2.$$

In particular, this implies $x_N > x - \varepsilon/2 = b + \varepsilon/2 > b$, leading to a contradiction.

(h) False. (The answer is True if (X, d_2) is a complete metric space.) Let $X = (0, 2) \subset \mathbb{R}$ inherit the metric d_2 from $\mathbb{R}$. Clearly, $S = (0, 2)$ is closed in X . Consider a sequence $x_n = 1/n$, $n \geq 1$ in S . Since $(x_n)_{n \geq 1}$ is convergent in $(\mathbb{R}, d_2)$, it is Cauchy in $(\mathbb{R}, d_2)$, and hence also Cauchy in S . However, $(x_n)_{n \geq 1}$ is not convergent in S since $x_n \rightarrow 0$ in $\mathbb{R}$, but $0 \notin S$.

2.

By the spectral theorem, it is diagonalizable in $\mathbb{R}$. The characteristic equation for A is

$$p(\lambda) = \lambda^2 - \text{tr}(A)\lambda + \det(A) = \lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1) = 0.$$

This yields $\lambda_1 = -1$ and $\lambda_2 = 3$. We see A has distinct real eigenvalues, so it is diagonalizable in $\mathbb{R}$.

We need to find eigenvectors $v_i, i \in \{1, 2\}$ which solves

$$(A - \lambda_i I)v_i = 0.$$

Such eigenvectors can be given as

$$v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

(Any multiples of v_1 and v_2 would also work.) It follows that for

$$P = [v_1 \ v_2] = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix},$$

it holds

$$AP = P \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix} \implies A = P \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix} P^{-1} = P\Lambda P^{-1}.$$

As a result, we have

$$\begin{aligned} A^n &= (P\Lambda P^{-1})^n = P\Lambda^n P^{-1} \\ &= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} (-1)^n & 0 \\ 0 & 3^n \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} (-1)^n + 3^n & (-1)^{n+1} + 3^n \\ (-1)^{n+1} + 3^n & (-1)^n + 3^n \end{bmatrix}. \end{aligned}$$

This shows A^n is not convergent as $n \rightarrow \infty$. (This is because some λ_i lies outside of $(-1, 1]$.) A is neither positive definite nor negative definite since λ_1 and λ_2 have different signs.

3.

(a) Yes, F is homogeneous. Let $\lambda > 0$, then

$$\begin{aligned} F(\lambda X_1, \lambda X_2, \dots, \lambda X_n) &= \sum_{i=1}^n \alpha_i (\lambda X_i)^\gamma \\ &= \lambda^\gamma \sum_{i=1}^n \alpha_i X_i^\gamma \\ &= \lambda^\gamma F(X_1, X_2, \dots, X_n). \end{aligned}$$

Thus, F is homogeneous of degree γ . Clearly, F is also homothetic.

(b) For notational convenience, allow $\gamma = 0$ for a moment. Let us introduce the notations

$$\begin{aligned} \Gamma_{qconv} &= \{\gamma \in \mathbb{R} : F \text{ is quasiconvex on } \mathbb{R}_{++}^n\}, \\ \Gamma_{qconc} &= \{\gamma \in \mathbb{R} : F \text{ is quasiconcave on } \mathbb{R}_{++}^n\}. \end{aligned}$$

We will verify that $\gamma \in \{0, 1\}$ are the knife-edge cases. I.e., for $n \geq 2$,

$$\Gamma_{qconv} = (-\infty, 0] \cup [1, \infty) \quad \text{and} \quad \Gamma_{qconc} = [0, 1].$$

If $n = 1$, then F is a monotone function regardless of γ , and hence

$$\Gamma_{qconv} = \Gamma_{qconc} = \mathbb{R}$$

as we noted from part (c) in question 1.

Let $g(t) := t^\gamma$ for $t > 0$. Then, $g''(t) = \gamma(\gamma - 1)t^{\gamma-2}$, so g is globally convex if $\gamma(\gamma - 1) \geq 0 \Leftrightarrow \gamma \in (-\infty, 0] \cup [1, \infty)$ and concave if $\gamma(\gamma - 1) \leq 0 \Leftrightarrow \gamma \in [0, 1]$. As a result, a positive linear combination

$$F(X_1, \dots, X_n) = \sum_{i=1}^n \alpha_i g(X_i)$$

of convex (concave) functions $(X_1, \dots, X_n) \mapsto g(X_i)$, $i \in \{1, 2, \dots, n\}$ is also convex (concave).

Alternatively, we could proceed by showing that

$$H_F(X_1, \dots, X_n) = \gamma(\gamma - 1) \begin{bmatrix} \alpha_1 X_1^{\gamma-2} & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \alpha_n X_n^{\gamma-2} \end{bmatrix}$$

is positive semi-definite and negative semi-definite for $\gamma \in (-\infty, 0] \cup [1, \infty)$ and $\gamma \in [0, 1]$, respectively, where H_F denotes the Hessian matrix of F .

Since convexity (concavity) implies quasiconvexity (quasiconcavity), we have estab-

lished one direction

$$(-\infty, 0] \cup [1, \infty) \subset \Gamma_{qconv}, \quad [0, 1] \subset \Gamma_{qconc}.$$

It remains to show that

$$(0, 1) \cap \Gamma_{qconv} = \emptyset, \quad (\mathbb{R} \setminus [0, 1]) \cap \Gamma_{qconc} = \emptyset.$$

This part is needed because quasiconvexity and quasiconcavity may not be naturally exclusive, i.e., a function can be quasiconvex and quasiconcave simultaneously even if it is not an affine function. (A point alluded by part (c) in question 1.) We instead argue that strict concavity (convexity) of F can rule out quasiconvexity (quasiconcavity).

Suppose that $\gamma \in (0, 1)$ and $F(X) = F(\tilde{X}) = 1$ for some $X = (X_1, \dots, X_n)$ and $\tilde{X} = (\tilde{X}_1, \dots, \tilde{X}_n)$ such that $X \neq \tilde{X}$. (There exist such X and $\tilde{X}$ if $n \geq 2$.) Then we can observe that

$$\begin{aligned} F\left(\frac{1}{2}X + \frac{1}{2}\tilde{X}\right) &= \sum_{i=1}^n \alpha_i \left(\frac{1}{2}X_i + \frac{1}{2}\tilde{X}_i\right)^\gamma \\ &> \sum_{i=1}^n \alpha_i \cdot \frac{1}{2} \left(X_i^\gamma + \tilde{X}_i^\gamma\right) \quad (\because \text{strict concavity of } g \text{ for } \gamma \in (0, 1)) \\ &= 1 = \max\{F(X), F(\tilde{X})\}, \end{aligned}$$

which contradicts quasiconvexity of F . This shows F is quasiconvex only if $\gamma \notin (0, 1)$. In a likewise manner, we can show that F is quasiconcave only if $\gamma \notin \mathbb{R} \setminus [0, 1]$. This establishes the rest:

$$\Gamma_{qconv} \subset (-\infty, 0] \cup [1, \infty), \quad \Gamma_{qconc} \subset [0, 1].$$

(c) Answer:

$$\hat{Y} = \sum_{i=1}^n \frac{\alpha_i \gamma (X_i^*)^\gamma}{\sum_{j=1}^n \alpha_j (X_j^*)^\gamma} \cdot \hat{X}_i.$$

Soln 1: We use two established facts: (1) if $y = \sum_{i=1}^n x_i$, then $\hat{y} = \sum_{i=1}^n \frac{x_i^*}{\sum_{j=1}^n x_j^*} \hat{x}_i$ and (2) $\widehat{ax^b} = b\hat{x}$ for $a, b \in \mathbb{R}$. Applying these rules to $Y = F(X_1, \dots, X_n)$, it follows

$$\begin{aligned} \hat{Y} &= \widehat{\sum_{i=1}^n \alpha_i X_i^\gamma} \\ &= \sum_{i=1}^n \frac{\alpha_i (X_i^*)^\gamma}{\sum_{j=1}^n \alpha_j (X_j^*)^\gamma} \widehat{\alpha_i X_i^\gamma} \\ &= \sum_{i=1}^n \frac{\alpha_i (X_i^*)^\gamma}{\sum_{j=1}^n \alpha_j (X_j^*)^\gamma} \gamma \hat{X}_i \end{aligned}$$

Soln 2: Let us denote $\Delta V := V - V^*$ for $V \in \{Y, X_1, \dots, X_n, X\}$. The first-order Taylor

expansion about $Y = F(X_1, \dots, X_n)$ around $(Y^*, X_1^*, \dots, X_n^*)$ gives us

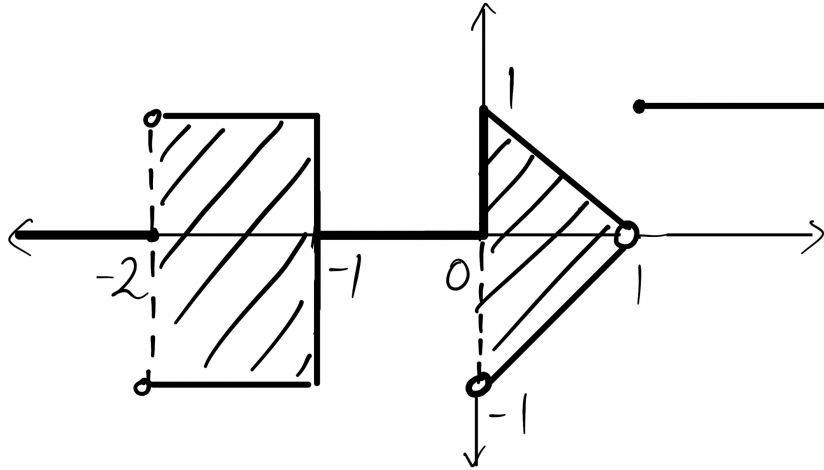
$$\begin{aligned}\Delta Y &= \nabla F(X_1^*, \dots, X_n^*) \cdot \Delta X \\ &= \sum_{i=1}^n \frac{\partial}{\partial X_i} F(X_1^*, \dots, X_n^*) \cdot \Delta X_i \\ &= \sum_{i=1}^n \alpha_i \gamma(X_i^*)^{\gamma-1} \Delta X_i.\end{aligned}$$

Dividing each side by $Y^* = \sum_{j=1}^n \alpha_j (X_j^*)^\gamma$, we obtain

$$\begin{aligned}\hat{Y} = \frac{\Delta Y}{Y^*} &= \sum_{i=1}^n \frac{\alpha_i \gamma(X_i^*)^{\gamma-1}}{\sum_{j=1}^n \alpha_j (X_j^*)^\gamma} \Delta X_i \\ &= \sum_{i=1}^n \frac{\alpha_i \gamma(X_i^*)^\gamma}{\sum_{j=1}^n \alpha_j (X_j^*)^\gamma} \frac{\Delta X_i}{X_i^*} \\ &= \sum_{i=1}^n \frac{\alpha_i \gamma(X_i^*)^\gamma}{\sum_{j=1}^n \alpha_j (X_j^*)^\gamma} \hat{X}_i.\end{aligned}$$

4.

(a)



(b) $\Gamma(\cdot)$ is upper hemi-continuous on

$$(-\infty, -2) \cup (-2, 0) \cup (0, 1) \cup (1, \infty) = \mathbb{R} \setminus \{-2, 0, 1\}$$

and lower hemi-continuous on

$$(-\infty, -1) \cup (-1, 0) \cup (0, 1) \cup (1, \infty) = \mathbb{R} \setminus \{-1, 0, 1\}.$$

5.

(a) Yes, a solution exists. The given constraints restrict the domain of the objective function to a compact set

$$B(p_1, p_2, m) := \{(x_1, x_2) \in \mathbb{R}^2 : p_1 x_1 + p_2 x_2 \leq m, x_1 \geq 0, x_2 \geq 0\}.$$

The set is bounded since $B(p_1, p_2, m) \subset [0, m/p_1] \times [0, m/p_2] \subset B_R(0)$ for $R = \max\{m/p_1, m/p_2\} + 1$ and closed as an inverse image of a closed set $(-\infty, m] \times \mathbb{R}_+^2$ by a continuous mapping $(x_1, x_2) \mapsto (p_1 x_1 + p_2 x_2, x_1, x_2)$. By Heine-Borel theorem, $B(p_1, p_2, m)$ is compact.

If $f(x_1, x_2) = \alpha \ln x_1 + \beta \ln x_2$ were continuous over $B(p_1, p_2, m)$, then we would have been able to conclude that a solution of the maximization problem exists. But Weierstrass theorem cannot be applied per se because f is not a well-defined real-valued function on $B(p_1, p_2, m)$. To circumvent this issue, we could transform the original problem by noting that maximizing f is equivalent to maximizing $\tilde{f}(x_1, x_2) = \exp(f(x_1, x_2)) = x_1^\alpha x_2^\beta$. Since $\tilde{f}$ is continuous on $B(p_1, p_2, m) \subset \mathbb{R}_+^2$, Weierstrass theorem implies that

$$\arg \max_{(x_1, x_2) \in B(p_1, p_2, m)} f(x_1, x_2) = \arg \max_{(x_1, x_2) \in B(p_1, p_2, m)} \tilde{f}(x_1, x_2) \neq \emptyset.$$

(b) Assume that the budget constraint does not bind at a maximum (x_1^*, x_2^*) . Let

$$\varepsilon = m - p_1 x_1^* - p_2 x_2^* > 0.$$

Then, $(x_1^* + \frac{\varepsilon}{2p_1}, x_2^* + \frac{\varepsilon}{2p_2})$ is feasible, but yields a strictly greater utility

$$f(x_1^* + \varepsilon/(2p_1), x_2^* + \varepsilon/(2p_2)) > f(x_1^*, x_2^*)$$

than a maximum (x_1^*, x_2^*) . This leads to a contradiction.

(c) Assume that the second constraint binds at a maximum (x_1^*, x_2^*) . Then

$$f(x_1^*, x_2^*) = \alpha \ln 0 + \beta \ln x_2^* = -\infty.$$

This is clearly not a maximum since $(\frac{m}{2p_1}, \frac{m}{2p_2}) \in B(p_1, p_2, m)$, while

$$f(m/(2p_1), m/(2p_2)) > -\infty.$$

Hence, the second constraint does not bind. By symmetry, the third constraint does not bind also.

(d) No. We have shown through part (b) and (c) that only the budget constraint is binding at a maximum. Since

$$\nabla(m - p_1x_1 - p_2x_2)(x_1^*, x_2^*) = -(p_1, p_2) \neq 0,$$

the constraint qualification is satisfied.

(e) Set up the Lagrangian as

$$\mathcal{L}(x_1, x_2, \lambda) = \alpha \ln x_1 + \beta \ln x_2 + \lambda(m - p_1x_1 - p_2x_2), \quad x_1, x_2 > 0, \lambda \geq 0.$$

We dropped the second and third constraints because they are not binding at a maximum. Then the first-order conditions are given as

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x_1} &= 0, & \frac{\alpha}{x_1^*} &= \lambda p_1, \\ \frac{\partial \mathcal{L}}{\partial x_2} &= 0, & \frac{\beta}{x_2^*} &= \lambda p_2, \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= 0, & m - p_1x_1^* - p_2x_2^* &= 0. \end{aligned}$$

Putting these together, we have

$$\begin{aligned} p_1x_1^* &= \frac{\alpha}{\lambda}, \\ p_2x_2^* &= \frac{\beta}{\lambda}, \\ m &= \frac{\alpha + \beta}{\lambda}, \end{aligned}$$

hence,

$$(x_1^*, x_2^*) = \left(\frac{\alpha m}{(\alpha + \beta)p_1}, \frac{\beta m}{(\alpha + \beta)p_2} \right).$$

(f) Let $\theta = (p_1, p_2, m, \alpha, \beta) \in \mathbb{R}_{++}^5$ and denote the utility function by $f(x, \theta)$ and the budget correspondence by $B(\theta) = B(p_1, p_2, m)$. Berge's theorem of maximum says that if $f(\cdot, \cdot)$ is continuous and $B(\cdot)$ is a non-empty-compact-valued and continuous correspondence, then

$$f^*(\theta) = \max_{x \in B(\theta)} f(x, \theta) \quad \text{and} \quad B^*(\theta) = \arg \max_{x \in B(\theta)} f(x, \theta)$$

are continuous and upper hemi-continuous in θ , respectively. The conditions on $f(\cdot, \cdot)$ and $B(\cdot)$ are all satisfied if we think of $\tilde{f}$ instead of f or restrict the domain of f to a proper compact subset of $B(\cdot)$ on which f is continuous. Since

$$B^*(\theta) = \{(x_1^*(\theta), x_2^*(\theta))\}$$

is a singleton set from part (e), upper hemi-continuity of $B^*(\theta)$ is equivalent to continuity of $\theta \mapsto (x_1^*(\theta), x_2^*(\theta))$.